Tab 1

Social Listening — Market Research Report

How Companies Build Social Listening Systems: Crawler, NLP, and LLM Layers

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1. Overview

Social listening systems used by companies — whether commercial platforms or custom-built — all share the same three-layer architecture. These layers are completely independent from each other:

Crawler Layer → Storage Layer → NLP/LLM Layer
"collect data" "store raw text" "understand data"

The crawler never does analysis. The NLP/LLM never visits websites. Each layer only communicates with the one next to it. This means any layer can be swapped out independently without affecting the others.

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2. The Crawler Layer — How Companies Collect Data

There are three ways companies collect social media and web data. Most platforms use a combination of all three.

2.1 Official Social Media APIs

Facebook, Twitter/X, Instagram, YouTube, and other platforms have official APIs — controlled channels that give structured data access. To use them at scale, a company must apply and be approved as an official partner.

This is a significant barrier — it costs money, requires legal agreements, and takes time to get approved. Zocial Eye, Mandala Analytics, and Zanroo are all registered Facebook Business Partners, which is why they can legally monitor Facebook public posts and comments at scale. This is not something a small team can simply set up overnight.

2.2 Custom Web Crawlers (Proprietary Crawlers)

For websites that do not have APIs — news sites, blogs, forums like Pantip — companies build their own web crawlers. "Proprietary" simply means they built it themselves and own it. It is the same concept as using Python `requests` + BeautifulSoup, but running at industrial scale.

Key differences from a basic scraper:

- Runs continuously 24/7 on cloud servers
- Handles thousands of URLs per hour automatically
- Manages IP rotation to avoid being blocked
- Handles cookie gates, JavaScript-rendered pages, and anti-bot systems
- Knows the specific HTML structure of target sites so it extracts clean data

Thai tools built their crawlers specifically for Thai websites, which is a major advantage over global tools. Their crawlers understand Pantip's comment structure, Thai news site layouts, and Thai cookie patterns out of the box.

What industry-level crawlers are built with

Scrapy — the most common Python framework for building production crawlers. It's what companies graduate to from BeautifulSoup when they need scale. BeautifulSoup just parses HTML — it has no scheduling, no queue management, no retry logic. Scrapy has all of that built in. It handles thousands of URLs, manages request queues, retries failed pages, and runs continuously.

BeautifulSoup = parses one page when you tell it to

Scrapy = manages crawling thousands of pages automatically

Playwright or Selenium — for JavaScript-rendered pages. These actually open a real browser in the background, execute the JavaScript, then extract the content. This is how you handle JS-heavy sites without paying Firecrawl.

Proxy rotation — not a single library but a system. They maintain pools of IP addresses and rotate between them on each request so websites can't detect and block them. Bright Data and Oxylabs actually sell this proxy infrastructure to other companies.

Custom scheduling system — runs crawlers on a schedule, like "check Pantip every 30 minutes for new posts mentioning these keywords." Built on top of tools like Celery (which KAM backend already uses) or Apache Airflow for larger scale.

2.3 Third-Party Data Providers

Some companies skip building their own crawler entirely and buy raw data feeds from providers like Bright Data or Data365. These providers already have crawlers running across thousands of social platforms and sell structured data as a service. This is a common shortcut for smaller platforms that want to focus on the analysis layer rather than data collection.

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3. The NLP/LLM Layer — How Companies Analyze Data

The NLP or LLM layer receives raw text from storage and produces structured analysis. Sentiment is just one small part of what it does. The full scope of analysis includes:

Analysis Type	What it does
Sentiment	Classifies text as positive, negative, or neutral
Topic categorization	What is this post about — claims, pricing, customer service, product launch
Intent detection	Is the person complaining, praising, asking a question, spreading news
Entity extraction	Pulls out brand names, product names, people mentioned from unstructured text
Language detection	Thai, English, or mixed — important for Thailand market
Emotion classification	Deeper than sentiment — angry vs sad vs disappointed vs happy
Trend detection	Looks across thousands of posts over time to detect when a topic is spiking

There are three approaches companies take for the NLP/LLM layer, from most difficult to easiest:

Approach 1 — Train From Scratch (old approach, rarely done now)

Build an entirely new model using custom labeled data. Requires massive datasets, years of work, and significant cost. No company building a new system today takes this approach.

Approach 2 — Fine-tune a Pretrained Model (current Thai tool approach)

Start with an existing pretrained language model and continue training it on your own domain-specific data. This is what Zocial Eye, Mandala, and Zanroo do.

The key model used in Thailand is **WangchanBERTa** — a Thai-language BERT model developed by NECTEC (Thailand's National Electronics and Computer Technology Center). It was pretrained on large amounts of Thai text, so it already understands Thai grammar, vocabulary, and structure.

Companies then fine-tune WangchanBERTa further on their own labeled datasets. For example, Wisesight collected and labeled 26,740 Thai social media messages as positive, neutral, negative, or question — and released this publicly as the Wisesight Sentiment Dataset. This dataset has been used extensively by researchers and companies to fine-tune Thai sentiment models.

The fine-tuning process means the model becomes specialized for their specific use case — Thai social media language, slang, industry-specific terms like insurance and banking vocabulary. That is what "in-house trained" means — they did not build the base model, but they trained it further with their own data which makes it more accurate for their context.

Approach 3 — Use an LLM Directly (modern approach)

Send text directly to a large language model like Gemini, GPT, or Claude with a structured prompt and receive analysis back. No training or fine-tuning required.

This is the approach used in this project. The tradeoff compared to fine-tuning is slightly higher cost per call and less consistency, but the quality is very competitive especially for Thai language since modern LLMs like Gemini already have strong Thai language capability built in. For a custom pipeline at this scale, using Gemini directly is faster to build, easier to maintain, and avoids the need for a dedicated ML training pipeline.

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4. Thailand-Specific Tools — Detailed Breakdown

4.1 Zocial Eye by Wisesight — Market Leader in Thailand

Who uses it: Enterprise companies across 20+ industries including banking, insurance, FMCG, and government. Currently the number one social analytics platform in Thailand with over 170 active clients.

Crawler layer:

- Official Facebook Business Partner — direct API access to Facebook public posts and comments
- Proprietary crawlers for Pantip, Thai news sites, blogs, and webboards built specifically for Thai site structures
- Continuous real-time data collection running on their own infrastructure

NLP/Analysis layer:

- Fine-tuned WangchanBERTa models trained on the Wisesight Sentiment Dataset
- Built-in AI system that continuously learns Thai slang and context over time
- Image recognition — detects brand logos in photos so it catches brand mentions even when no text is written
- OCR (Optical Character Recognition) — reads text embedded inside images, for example a photo of a complaint letter or a screenshot of a review

How companies use it: A marketing or PR team subscribes to Zocial Eye, sets their keywords — brand name, competitor names, product terms. The platform continuously collects and analyzes matching content. The team logs into a dashboard to see sentiment trends, topic breakdowns, and can generate one-click campaign reports. No technical work required from the client side.

4.2 Mandala Analytics — Mid-Range, Most Accessible

Who uses it: Mid-size brands entering Thailand, marketing agencies, companies that want social listening without needing a specialist analyst to interpret results.

Crawler layer:

- Official social media API partnerships similar to Zocial Eye
- Custom crawlers covering Thai web sources
- Importantly includes Pantip via a dedicated Forum Listening feature — critical for Thailand since Pantip is where Thai consumers write detailed complaints about insurance, banks, and hospitals

NLP/Analysis layer:

- Proprietary Thai NLP pipeline focused on producing clean, easy-to-read outputs without requiring analyst expertise
- Strong visualization layer — results are automatically turned into charts, trend graphs, and campaign summaries inside the dashboard

How companies use it: More self-serve than Zocial Eye. A brand manager can set up keyword tracking and read the dashboard results without needing a data analyst to interpret the numbers. Recommended for companies that are new to social listening and want results quickly without a steep learning curve.

4.3 Zanroo — Most Technical, ASEAN-Focused

Who uses it: Enterprise clients in insurance, telecommunications, banking, FMCG, and logistics across Southeast Asia. Based in Nonthaburi, Thailand.

Crawler layer:

- Proprietary data collection using a technology they call Information Retrieval (IR)
- Extracts not just text but structured metadata — location of the post, detected emotion category, language percentage (how much Thai vs English in a single post), objects detected in images
- Real-time data collection optimized for crisis monitoring — faster update cycle than standard crawlers

NLP/Analysis layer:

- Multi-language NLP supporting Thai, Mandarin, Bahasa, Japanese, Tagalog, Vietnamese, and Burmese — fine-tuned separately for each language
- Goes beyond sentiment into full social analytics: influencer identification and scoring, topic clustering, predictive analysis
- Unique feature: connects the NLP analysis layer directly into a customer engagement system — so a complaint detected by the crawler automatically becomes a customer support ticket routed to the right team

How companies use it: Zanroo is used differently from the other two. Beyond just monitoring, their system integrates listening data into internal workflows. An insurance company using Zanroo might have it automatically detect a complaint about claim delays on social media and create a support ticket in their CRM before the customer even contacts them directly.

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5. Global Tools vs Thai Tools — Key Difference

	Zocial Eye / Mandala / Zanroo	Brandwatch / Talkwalker / Meltwater
Thai language NLP	Native, trained on Thai data	Weak — needs human analyst on top
Pantip coverage	Yes — proprietary crawler	No
Facebook access	Official Business Partner	Official partnerships but less localized
Pricing	Lower, local market pricing	$6,000–$100,000+ per year
Best for	Thai market focused brands	Global/regional enterprise programs

Global tools like Talkwalker are used by multinational companies like AXA and AIA at the group level for global brand monitoring. For Thailand-specific consumer intelligence, those same companies would still need a local tool like Zocial Eye on top of it.

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6. Summary — What Approach Fits This Project

Based on the market research above, the approach that makes sense for AIBL's custom pipeline is:

Layer	Industry standard	This project
Crawler	Proprietary crawler + official API	BeautifulSoup (news/corporate sites) + Gemini url_context fallback

NLP/LLM	Fine-tuned WangchanBERTa	Gemini 2.5 Flash via Vertex AI
Storage	Internal database + dashboard	Notion (POC) → KAM database (production)

The crawler layer gap compared to Zocial Eye is Facebook and Pantip — both require official API partnership or a third-party data provider. This is a decision point for production phase.

The NLP layer using Gemini directly is actually the more modern approach compared to fine-tuning. For this project scope it is faster to build, easier to maintain, handles both Thai and English in one model, and produces high quality analysis without a dedicated ML training pipeline.

Tab 2

Two totally separate problems:

Problem 1: How do you get the text from a website?
Problem 2: How do you get data from social media platforms?

These are completely different challenges. BeautifulSoup and Gemini only solve Problem 1.

Problem 1 — Websites (news sites, corporate sites, blogs)

This is what you've been doing. AXA, ThaiLife, AIA, Viriyah — these are all normal websites. For these:

BeautifulSoup → works for simple static sites
Firecrawl → works for JavaScript-heavy sites (like AXA)
Gemini url_context → quick fallback but gets blocked by WAF

When WAF happens: WAF blocks requests that look like bots. The pattern from your test:

- Gemini url_context fetches from Google cloud servers → looks like a bot → WAF blocks it
- BeautifulSoup from your local machine → looks like a normal browser → gets through
- BeautifulSoup from a cloud server (like Google Cloud Run) → might get blocked too
- Firecrawl handles WAF automatically — it has built-in IP rotation and proxy management, that's part of what you pay for

So for websites the decision tree is simple:

Try BeautifulSoup first
  → if JS-heavy (content doesn't load) → use Firecrawl
  → if WAF blocks cloud deployment → run BeautifulSoup locally
    OR pay Firecrawl to handle it

For your current scope — targeting specific insurance company websites — BeautifulSoup + Firecrawl as backup covers you completely. No third-party provider needed.

Problem 2 — Social Media Platforms (Facebook, Pantip, Instagram, Twitter)

This is completely different and this is where it gets complicated.

Facebook, Instagram, Twitter/X — you **cannot** just scrape these with BeautifulSoup or Firecrawl. They have:

- Login walls (content only visible if you're logged in)
- Heavy JavaScript rendering
- Active anti-bot systems that detect and ban scrapers
- Legal terms that prohibit scraping

To get data from social media you have three options:

Option A — Official API Partnership Apply directly to Facebook/Meta, Twitter/X, etc. to become an approved data partner. This is what Zocial Eye, Mandala, Zanroo did.

- Takes months to get approved
- Costs money for API access tiers
- Legal and compliant
- Gets you clean structured data directly from the platform

Option B — Third-Party Data Provider Pay a company like Bright Data, Apify, or Data365 who already has the API partnerships or infrastructure to collect social media data. They sell you the data via their API.

- You pay monthly and call their API
- They handle all the collection complexity
- Faster to set up than Option A
- Still costs money

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the Workflow

Workflow A — Current setup (BeautifulSoup + Gemini)

Your code visits website
→ BeautifulSoup extracts text
→ Send text to Gemini
→ Gemini returns sentiment/topics/summary
→ Save to Notion/database

Covers: Corporate websites, news sites, blogs Does NOT cover: Facebook posts, Pantip threads, Instagram, Twitter

Workflow B — Add Firecrawl

Your code calls Firecrawl API with a URL
→ Firecrawl visits the site (handles JS, WAF, proxies)
→ Returns clean text to your code
→ Send text to Gemini
→ Gemini returns analysis

→ Save to database

Covers: Everything BeautifulSoup covers + JavaScript-heavy sites + WAF-protected sites Still does NOT cover: Social media platforms

Workflow C — Add Third-Party Data Provider (e.g. Apify)

Your code calls Apify API: "give me Facebook posts mentioning AXA Thailand"
→ Apify collects the posts from Facebook
→ Returns structured data to your code
→ Send text to Gemini
→ Gemini returns analysis
→ Save to database

Covers: Social media posts and comments You still need BeautifulSoup/Firecrawl separately for websites — Apify covers social media, not corporate websites

Workflow D — Use Zocial Eye or Mandala (full SaaS tool)

You log into Zocial Eye dashboard
→ Set your keywords
→ Zocial Eye handles EVERYTHING (crawling websites + social media)
→ Shows you results in their dashboard

You don't write any code. You don't need BeautifulSoup, Firecrawl, Gemini, or any provider. Everything is inside their platform already. But you can't connect it to KAM directly — you'd have to manually export reports.